

A Continuous Indian Sign Language Production System from Isolated Signs

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Abstract—Indian Sign Language (ISL) is a low-resource language, and the scarcity of well-annotated continuous datasets significantly restricts the development of robust applications for sign language recognition, translation, and production. The lack of such resources limits research advancements and the deployment of real-world ISL-based technologies. The primary objectives of this paper are: 1) To develop a sign language production (SLP) system that translates spoken language sentences into Indian Sign Language while incorporating proper co-articulation for natural and fluent signing, 2) To create a Synthetic Indian Continuous Sign Language (SynICS) dataset, which can serve as a valuable resource for future ISL translation research. To evaluate the effectiveness of our approach, we employ an off-the-shelf sign language translation (SLT) model for back-translation, assessing whether the synthetic signs can be reliably converted back into meaningful text. We train this model on the SynICS dataset and evaluate real-life videos from the iSign dataset. Our results are highly promising, achieving a BLEU-4 score of 17.56, with the translated sentences semantically matching the ground truth. These findings demonstrate the potential of synthetic continuous ISL data, generated from isolated signs, to improve Indian SLT task. The SLP system and the generated dataset will be publicly available on https://cs.rkmvu.ac.in/~isl/sl_gen.

Index Terms—Indian sign language, sign language production, sign language translation, sign language dataset

I. INTRODUCTION

Sign languages are created and utilized by the deaf community to facilitate communication. These are visual languages; analogous to phonemes in spoken language, sign languages use hand gestures, facial expressions, and body movements. Sign languages have unique grammatical structures and linguistic complexity, very different from spoken languages [17]. To bridge the communication gap between hearing people and the deaf community, converting sign language to spoken language and vice-versa is required.

Sign Language Translation (SLT) converts sign language (SL) to spoken language, while Sign Language Production (SLP) does the reverse. Both have two main approaches: 1) gloss-driven: uses gloss¹ sequences as an intermediate step [2], [24], [27], and 2) end-to-end: directly translates between SL videos and spoken language [2], [18]. Most of the works in these two tasks are based on the Czech Sign Language dataset RWTH-PHOENIXWeather-2014T (PHOENIX14T) [2].

¹Glosses are written representations of sign language units.

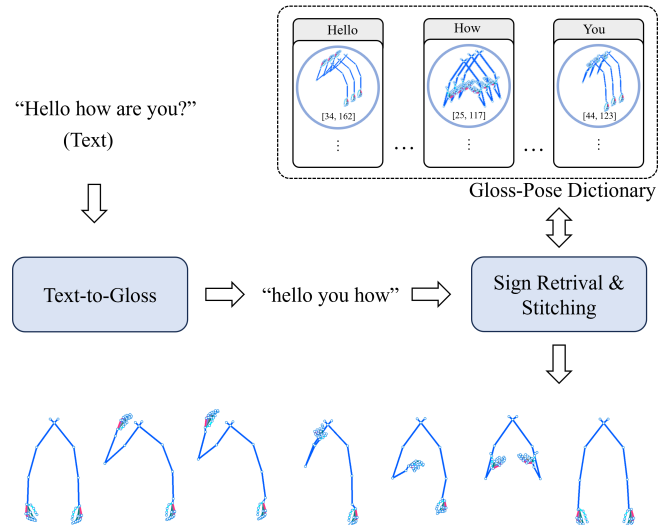


Fig. 1. Overview of our Sign Language Production system. Each input sentence is passed through the Text-to-Gloss algorithm to generate the glosses. Then a pose sequence for each gloss is retrieved from the Sign-Pose dictionary and stitched together to generate a co-articulated continuous sign.

Despite being the most used SL in the world, very little research has been conducted on India Sign Language². Some approaches [11], [21] utilize HamNoSys [8] notation based systems to tackle the Indian SLP task. However, the generated avatar animations are not natural and not co-articulated. In the case of the Indian SLT task, recently Joshi *et al.* [9] proposed a dataset called iSign, having videos with their corresponding English text, which were collected from different web sources. This dataset doesn't contain gloss sequences, which restricts us from applying gloss-driven SLT approaches. Due to various limitations, end-to-end SLT approaches perform poorly on this dataset.

The aforementioned challenges encourage us to build a combined pipeline that tackles both SLP and SLT tasks for Indian sign language. We follow a keypoints-based gloss-driven approach where we first build an SLP system that produces natural co-articulated continuous SL videos. We generate a synthetic continuous SL dataset using this system, which is used to train a gloss-driven SLT model for translation. Since

²<https://www.ethnologue.com/guides/ethnologue200>

no datasets with gloss annotations are available for the Indian sign language, we employ a deterministic algorithm based on ISL grammatical rules proposed by Sugandhi *et al.* [21] to translate English sentences to gloss sequence. We create a dictionary from a large isolated SL dataset called FDMSE-ISL [15]. Each gloss entry in this dictionary maps to multiple 3D pose sequences, extracted from isolated sign videos. For a given gloss sequence, we retrieve the corresponding pose sequences from the dictionary and stitch them using linear interpolation to incorporate co-articulation. Using this SLP system, we generate a synthetic sign language dataset from a Hindi movie subtitle corpus OpenSubtitles³. We train a gloss-driven Transformer-based Sign Language Translation (SLT) model [3] using the synthetic continuous sign language dataset. To evaluate the effectiveness of our approach, we test the model on real-life samples from the iSign dataset [9].

In summary, the contributions of this work are listed below,

- We propose a gloss-driven dictionary-based SLP system that generates continuous signs with co-articulation for Indian sign language.
- We present a synthetic continuous Indian sign language dataset with gloss annotations.
- We propose a sign language translation pipeline based on an isolated sign language dataset. To the best of our knowledge, this is the first approach to utilize synthetic data for training and real-life data for testing in the sign language translation task.

II. RELATED WORKS

Sign Language Translation: Automatic Sign Language Translation (SLT) using computational methods has been studied for the last three decades [23]. Initially, researchers tackled the Continuous Sign Language Recognition (CSLR) problem using the Hidden Markov Model [19], [26] and Artificial Neural Network [5]. Kollar *et al.* [10] proposed a statistical system for CSLR, integrating hand and face tracking with both visual and language modeling. Nair *et al.* [14] have proposed a Neural Machine Translation (NMT) method for Indian CSLR task using BiLSTM. Camgoz *et al.* [2] first introduced the Sign-to-Text (S2T) task in the framework of NMT using Recurrent Neural Network (RNN) [22]. Subsequently, a gloss-driven SLT approach [3] is proposed with gloss sequence generation as an intermediate step using the Transformer [25] model. Since then this architecture has become the standard for quantitative evaluation (back-translation) of Sign Language Production (SLP) methods.

Sign Language Production: Initial Sign Language Production (SLP) approaches [4], [6], [13] generated avatar animations from motion capture data. Dependency on motion capture suits and equipment makes these approaches expensive to scale. After the development of the Hamburg Notation System (HamNoSys) [8], researchers leveraged it to generate avatar-based sign language animations [1], [21], [28]. However, these systems rely on HamNoSys experts for adaptation to

regional languages, and the robotic nature of the generated signs remains unpopular within the Deaf community.

Recently, Stoll *et al.* [20] proposed a gloss-driven Sign Language Production (SLP) system that generates RGB sign language videos. The system operates translated text-to-gloss using Recurrent Neural Network (RNN) [22] and use a gloss-to-pose dictionary and generates pose-to-video using Generative Adversarial Network (GAN) [7]. However, this approach has notable limitations such as missing finger details, distorted hand shapes, and temporal inconsistencies. Zuo *et al.* [27] segmented continuous sign videos and generated isolated sign videos. Then they applied a variant of SMPLify-X [16] to generate 3D avatar signs for each gloss and created a gloss-3D dictionary. Given a text input, they first translate it into a gloss sequence using machine translation. Then, they stitch the corresponding isolated 3D signs retrieved from the dictionary, applying interpolation to ensure smooth co-articulation. Although this approach enhances sign language understanding by generating 3D avatar animations, storing animations for all isolated signs results in a significant memory overhead.

We follow a similar gloss-driven dictionary-based approach, where we create a gloss-3D pose sequence dictionary. We first translate gloss sequence from text using a deterministic Indian sign language grammatical rule based algorithm proposed by Sugandhi *et al.* [21]. We retrieve the corresponding isolated pose sequences from the dictionary and stitch them by applying interpolation in between for co-articulation. This method is easily scalable by simply updating the dictionary and minimal computation make the generation process real-time.

III. METHODOLOGY

In this section, we provide a detailed methodology of our Sign Language Production (SLP) system in (Section III-A).

A. Sign Language Production System

The gloss-driven SLP system generally follows the following formulation: given a text $X = [x_1, x_2, \dots, x_W]$ containing W words, it is first translated to a gloss sequence $L = [l_1, l_2, \dots, l_G]$ of G glosses. For all the gloss in the gloss sequence, corresponding video or pose sequences are retrieved from a dictionary. Finally, from these pose sequences an output video, pose sequence, or avatar animation $Y = [y_1, y_2, \dots, y_O]$ is generated. Fig. 1 illustrates our overall SLP system.

Gloss-Pose dictionary: Our SLP system utilizes a large-scale Indian isolated Sign Language dataset, FDMSE-ISL [15], to construct the gloss-to-3D pose sequence dictionary. FDMSE-ISL dataset contains $M = 2002$ glosses with 20 video samples per gloss. We extract 3D keypoints from all sample videos using the MediaPipe Holistic library [12]. Each frame yields 543 3D keypoints, including those for the face, pose, left hand, and right hand. However, we primarily focus on arm keypoints, as they contribute the most to Indian Sign Language. So we select $K = 51$ keypoints (21 per hand, 2 eyes, 1 nose, 2 shoulders, 2 forearms, 2 wrists) for each frame. In the dataset, there are multiple signers, so keypoint normalization is required. Since the keypoints are in the

³<https://opus.nlpl.eu/OpenSubtitles/hi&en/v2024/OpenSubtitles>

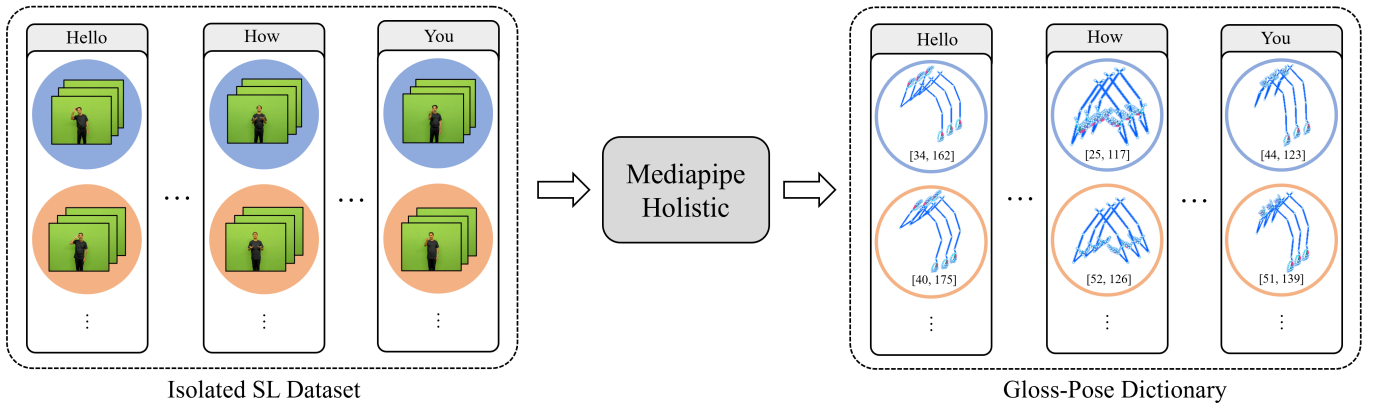


Fig. 2. Creation of a gloss-Pose dictionary from isolated sign dataset. For each gloss, we extract pose sequences from video samples performed by different signers. In the gloss-pose dictionary, each gloss is mapped to its associated pose sequences and their signing start and end frame indices.

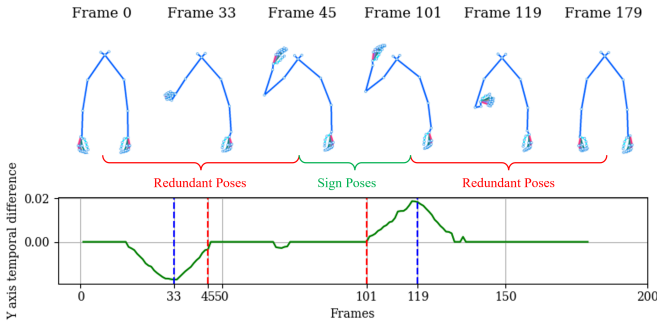


Fig. 3. Plot of y axis temporal difference of dominant hand (right) wrist keypoint from a sample pose sequence of the gloss “Hello” in the FDMSE-ISL [15] dataset. The blue lines indicate the starting and ending cross indices, while the red lines indicate the starting and ending indices of the sign.

$[0, 1]$ range, we first project them onto the image plane by multiplying the x and y coordinates by the frame’s width and height, respectively. Additionally, since the z coordinate shares the same scale as the x -axis⁴ we multiply it by the frame’s width. Next, we compute the Euclidean distance between the two shoulder keypoints and use it as the unit distance. We then define a square bounding box centered at the nose keypoint with a side length of six times the unit distance. Finally, we normalize all keypoints by scaling them with the side length. Fig. 2 illustrates the Gloss-Pose dictionary creation process..

After extracting and normalizing the keypoints from all videos, we construct a dictionary where each gloss g_i maps to a list of pose sequences $[P_i^1, P_i^2, \dots, P_i^S]$, where P_i^s represents the 3D pose sequence of subject s . Furthermore, P_i^s consists of $[S_{i,1}^s, S_{i,2}^s, \dots, S_{i,F}^s]$, where $S_{i,f}^s \in \mathbb{R}^{K \times 3}$ denotes the pose at frame f and F varies for each video instance. In the FDMSE-ISL [15] dataset, the signers start and end in a rest pose, and these redundant poses need to be removed. To achieve this, we apply a simple algorithm based on the prior knowledge that signs in Indian Sign Language are typically performed at the chest level. We compute the y -axis temporal difference by

subtracting the y -component of the wrist keypoint of frame f from the y -component of the wrist keypoint of frame $f + 1$. We identify the starting and ending cross indices where the y value of the wrist keypoint equals the y value of the elbow keypoint. From the starting cross index, we move forward, and from the ending cross index, we move backwards, stopping where the y -axis temporal difference is close to 0 (we select a threshold of 0.002). This process helps us to determine the frame indices within which the sign occurs. Fig. 3 illustrates the y -axis temporal difference of the wrist keypoint for a sample video, highlighting the cross indices as well as the signing start and end indices. In the gloss-pose dictionary, we also keep these indices I_i^s corresponding to the pose sequence P_i^s for trimming.

Text-to-Gloss: We have to translate English text to Indian Sign Language (ISL) glosses. Inspired by the ISL grammatical rules proposed by Sugandhi *et al.* [21] we develop an algorithm. To parse a sentence and extract parts-of-speech (POS) tags and other morphological information about the words we use the Stanza library⁵. The ISL gloss sequences follow Subject-Object-Verb (SOV) ordering, for this, we take the root word and traverse the semantic tree recursively and change the ordering. Next, we remove punctuation, adjectives, articles, prepositions, auxiliary verbs, conjunctions, etc. Additionally, we decompose the abbreviations, and proper nouns into alphabets for finger-spelling. Finally, the remaining words are the ISL gloss sequence.

Sign Retrieval and Stitching: After the translation from text to ISL glosses, we get a gloss sequence $L = [l_1, l_2, \dots, l_G]$. For a gloss l_k , we find the corresponding gloss g_i from the FDMSE-ISL vocabulary. Next, we retrieve the corresponding 3D pose sequence P_i^s and sign indices I_i^s from the gloss-pose dictionary, where we choose the subject s randomly. We trim the pose sequence by removing the redundant indices from start to $I_i^s[0]$ and $I_i^s[1]$ to end. After processing all the glosses we get the pose sequences $[Q_1, Q_2, \dots, Q_G]$. A naive approach, as used by Sugandhi *et al.* [21], is to concatenate

⁴<https://mediapipe.readthedocs.io/en/latest/solutions/hands.html>

⁵<https://stanfordnlp.github.io/stanza/>

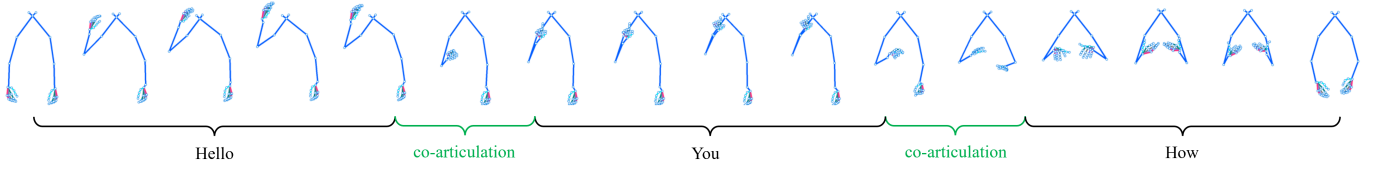


Fig. 4. The pose sequence for the sentence “Hello, how are you?” from glosses [“Hello”, “You”, “How”] stitched together with co-articulated pose sequences in between each isolated sign.

TABLE I
COMPARISON OF SAMPLE SENTENCES PRESENT IN THE iSIGN [9]
DATASET AND THE TRANSLATED SENTENCES GENERATED BY SLT [3]
MODEL TRAINED ON THE SYNICS DATASET. BLUE AND RED COLORED
TEXT HIGHLIGHT THE DIFFERENCE BETWEEN iSIGN [9] DATASET
SENTENCES AND THE TRANSLATED SENTENCES.

iSign [9] Dataset Sentence	Translated Sentence
these are my eyes	this is my eyes
where was haldi going	where are they going
he was a doll c	he has a family
her name is moni	and her name is mary
3 how old are you	how old are you here
5 what is this	what is this call
and so many cats	so many stars
where are you going to	where are you going
how did they know this	how do they know this
how do you feel	how would you feel now
how many people live in your house	how many people are in your house
here is a picture of my home	this is in a picture of here
4 give me a handkerchief	give me the city
i was happy to do that	im happy to do it
what do you find	what are you find
how did they know this	how do they know this
my father is a farmer	my father is get me
my sister and my brother	and my sister and my brother
this is your page	this is your friends
but while he was doing this	but if he do this

the sequences along the time dimension. However, this results in discontinuous pose sequences that are visually not pleasing and show poor generalization in the SLT task. So, we have to generate the co-articulations between adjacent pose sequences. We take the last pose of Q_{g-1} and the first pose of Q_g ($2 \leq g \leq G$) and linearly interpolate between them with a fixed duration of n frames to generate the co-articulation C_g^{g-1} , where C_g^{g-1} is the interpolated pose sequence of duration n . Finally the modified pose sequences become $[Q_1, C_2^1, Q_2, C_3^2, Q_3, \dots, Q_{G-1}, C_G^{G-1}, Q_G]$. An example complete output pose sequence is shown in Fig. 4. This SLP system is modular, which makes it easy to extend to a larger vocabulary by adding new isolated signs to the gloss-pose dictionary without requiring architectural changes.

IV. SYNTHETIC INDIAN CONTINUOUS SIGN LANGUAGE (SYNICS) DATASET

In the context of Indian Sign Language (ISL), there exists only one large-scale continuous sign language (SL) dataset called iSign [9]. This dataset contains sign videos scraped from multiple sources. A major drawback of this dataset is that

it performs poorly on sign language translation (SLT) tasks due to the absence of gloss annotations, dataset noise (miss-labelling) and inconsistent grammar rules across different sign videos. Developing a well-annotated continuous SL dataset is both a tedious and costly process. To address this problem, we propose an alternative approach by creating a synthetic continuous ISL dataset to provide resources for the Indian SLT task.

To develop such a dataset, we first require a corpus containing sentences relevant to the Indian context. For this, we select a Hindi movie subtitle corpus from OpenSubtitles⁶. This corpus contains ≈ 3 million Hindi sentences and their corresponding English translations. We randomly select a subset of 30,000 sentences to ensure our computational feasibility. Next, we apply our Text-to-Gloss algorithm to translate the sentences into ISL glosses. We then filter these sentences based on the availability of their glosses in the FDMSE-ISL [15] dataset vocabulary. Finally, we obtain 4,125 sentences, which we split into training (2,951), validation (387), and test (787) sets, ensuring that all glosses in the validation and test sets are present in the training set. Using our SLP process, we generate synthetic continuous signs for all sentences. During the sign retrieval and stitching process for the train, test, or validation set, we randomly select pose sequences from the corresponding set in the FDMSE-ISL [15] dataset to enhance the dataset’s generalizability. Finally, each sample of this dataset contains English text, English gloss sequence, and pose sequence of corresponding glosses.

V. EXPERIMENTS AND RESULTS

A. Baseline Model

To evaluate the effectiveness of the proposed synthetic data generation framework, we chose to use the Sign Language Transformers [3] model. This model is used as a benchmark across different continuous sign language translation works and in obtaining the baseline machine translation results for the iSign [9] dataset. The transformer encoder in this model takes as input the feature matrix for all frames in a sign video. The 3D pose sequence in $\mathbb{R}^{F \times K \times 3}$ of each sign sentence in the SynICS dataset is flattened in $\mathbb{R}^{F \times (K \times 3)}$ and passed as input to the transformer encoder.

B. Test Dataset Selection

To evaluate our proposed SLP system, we select a subset of 100 samples from the iSign [9] dataset. These samples are

⁶<https://opus.nlpl.eu/OpenSubtitles/hi&en/v2024/OpenSubtitles>

TABLE II
MACHINE TRANSLATION RESULTS ON THE SYNICS VALIDATION SET, TEST SET AND ON THE ISign [9] SUBSET DATASET. $\uparrow$ INDICATES BIGGER NUMBER IS BETTER WHEREAS $\downarrow$ INDICATES SMALLER NUMBER IS BETTER

Dataset	Stitching Method	Word Error Rate $\downarrow$	BLEU-1 $\uparrow$	BLEU-2 $\uparrow$	BLEU-3 $\uparrow$	BLEU-4 $\uparrow$	CHRF $\uparrow$	ROUGE $\uparrow$
SynICS (Dev Set)	Concatenation	69.67	29.19	16.93	11.95	8.92	23.77	29.84
SynICS (Test Set)	Concatenation	79.25	24.67	12.56	7.75	5.17	20.08	24.21
iSign Subset	-	68.10	35.29	22.61	15.87	11.35	27.95	36.31
SynICS (Dev Set)	Interpolation	58.20	29.52	17.31	11.96	9.04	24.63	29.60
SynICS (Test Set)	Interpolation	70.87	23.78	12.49	7.59	5.05	19.82	24.09
iSign Subset	-	55.70	45.91	33.26	24.74	17.56	36.67	47.18

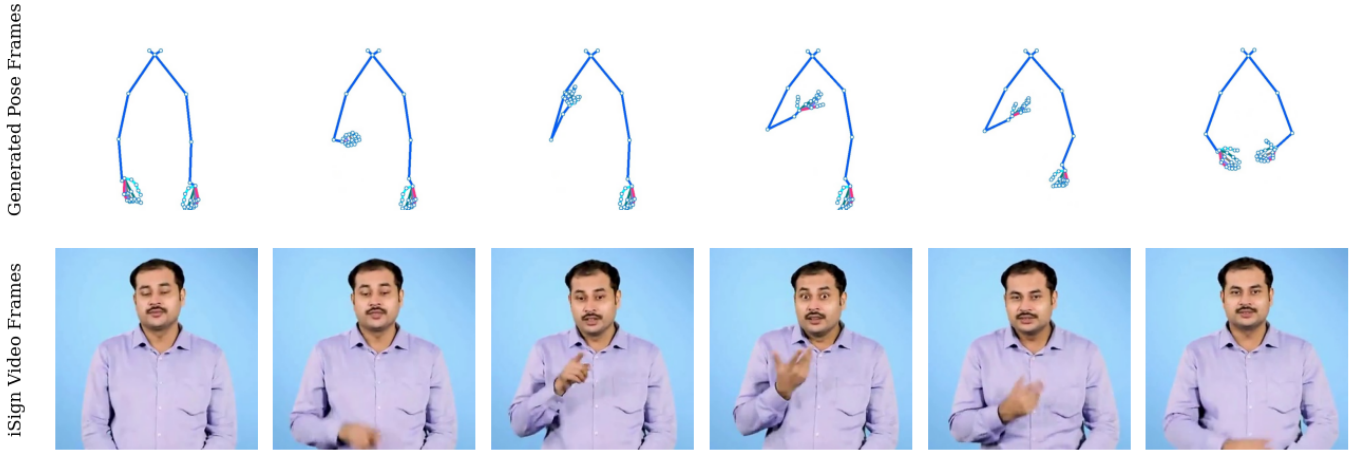


Fig. 5. The generated pose sequence for the sentence “How do you feel?” compared with the real sign video from the iSign [9] subset dataset.

based on the sign similarity of the constituent isolated signs in the sample sentences from the iSign [9] dataset, which are also present in the FDMSE-ISL [15] dataset. Mediapipe-Holistic [12] image-to-pose model is used to obtain the 3D pose sequence for each selected sign video. This real-life test set is used to validate the generalization ability of our approach, demonstrating that synthetic data can be effectively used for training sign language translation models to translate real sign videos into English sentences.

C. Evaluation of Proposed Sign Language Production System

To gauge the effectiveness of the proposed SLP system, we present both quantitative and qualitative results.

1) *Quantitative Results:* To evaluate any machine translation performance, several metrics like the Word Error Rate (WER), BLEU-1, BLEU-2, BLEU-3, BLEU-4, Character F-score (CHRF) and ROUGE are used. The quality of machine translation results on the SynICS validation set, test set, and the iSign [9] subset dataset is evaluated using these seven different translation metrics as shown in Table II. We compare the results for synthetic data generated by two different stitching methods, concatenation and interpolation. The SLT model [2] trained on the synthetic data generated using the interpolation stitching technique shows better performance on real data. The relatively high BLEU-4 and ROUGE score and the low WER of 17.56, 47.18 and 55.70 respectively indicates

good performance on the iSign [9] subset dataset. This further indicates a good translation performance of the SLT model [3] on real data when trained with synthetic data generated by the proposed Sign Language Production system.

2) *Qualitative Results:* To evaluate the quality of the SLT [3] model trained on the SynICS dataset we compare the sentences generated for any pose sequence input from the iSign [9] subset dataset with the actual sentences. Twenty such sentence pairs are shown in Table I. The model generally produces accurate sentences, though it occasionally struggles with correctly predicting names or verb forms. The tenses of some predicted sentences are also different but overall the model succeeds in translating any sign sequence to accurate and meaningful English sentence.

Another qualitative measure is visually judging the output pose sequence generated by the SLP system for any input sentence. In Fig. 5, a frame-by-frame comparison between the generated pose sequence samples and the real video frame samples is presented. The generated pose sequence is very similar to the actual video with pose of hand sequence closely resembling the hand movement in the video.

VI. CONCLUSION

In conclusion, we introduce a Sign Language Production (SLP) system, capable of generating continuous sign sequences from isolated signs in real-time, with minimal

computation cost. Additionally, we present a synthetic Indian continuous Sign Language dataset, SynICS, which serves as a benchmark for validating this SLP system. This SynICS dataset was used to train a sign language translation model that was then tested on a real continuous sign language dataset. Our results demonstrate that an existing sign language translation model trained on the synthetic dataset generated by this SLP system performs effectively in translating real sign videos. This innovative system addresses the shortage of continuous Indian Sign Language datasets by enabling the creation of synthetic continuous datasets customized for specific applications using readily available isolated sign language datasets.

VII. LIMITATIONS

Our SLP system relies on a gloss-pose dictionary, which limits its ability to handle unseen glosses without adding new isolated sign samples. Additionally, while synthetic data enables scalable training, it may not fully capture the nuances of real-world signing such as non-manual signals or signer-specific variations.

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